

Full-length Article

Searching for blood biomarkers and treatment targets in Women with fibromyalgia – Protein interaction patterns and anti-satellite glia cell IgG antibodies as promising candidates

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ABSTRACT

Background: Recent studies suggest that autoreactive immunoglobulin G (IgG) antibodies binding to satellite glia cells (anti-SGC IgG) in the dorsal root ganglia influence pain intensity in a subgroup of fibromyalgia subjects (FMS), thus indicating altered immune activation. The main aim of this study was to identify proteins distinguishing female FMS from female healthy controls (HC) and within the FM group, proteins distinguishing FMS with high vs low levels of anti-SGC IgG. The secondary aim was to assess the associations between serum proteins and anti-SGC IgG, respectively, and FM symptoms.

Methods: Anti-SGC IgG was quantified using an immunofluorescence assay. Proteins in serum were assessed using Olink® Explore 384 Inflammation panel, regarding differences between FMS (n = 93) and HC (n = 40) and regarding differences between FMS with high (≥50 %) and low (<50 %) anti-SGC IgG, respectively. Proteins found to differ between groups (VIP ≥ 1.3) were further analyzed regarding protein-interactions using the software tool STRING (FM vs HC n = 56, high vs low anti-SGC IgG n = 55). Results from the FM group were also compared with two nociceptive pain conditions.

Results: In FMS, a cluster of immune system-related proteins was found among upregulated proteins, including CD40 and CD40L, with central roles in humoral immune response. CD40 levels were associated with more severe FM symptoms. In contrast, a cluster of tissue development-/regeneration-related proteins was found among downregulated proteins, this was not seen in nociceptive pain conditions. In FMS with high anti-SGC IgG, clusters dominated by immune system-related proteins were found among both upregulated and downregulated proteins. The cluster of upregulated proteins included CD79b, a protein necessary for B-cell receptor function, and CD4, a co receptor needed for T cell activation, thus with central role in activating various immune responses, including B-cell activation. Positive correlations were seen between some of these proteins and symptoms. On the contrary, several of the downregulated proteins correlated negatively to symptoms.

Conclusion: Our data support the involvement of the immune system in FM and indicate that further studies on autoimmune mechanisms, proteomics, and protein interaction analysis could lead to new objective diagnostic criteria identifying FMS likely to benefit from immunomodulatory treatments.

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1. Introduction

Fibromyalgia (FM) is a chronic pain syndrome with considerable impact on function and quality of life (Bucourt et al., 2021). Typical symptoms are chronic widespread pain, tenderness, fatigue, non-refreshing sleep, cognitive symptoms and mood disturbances (Wolfe et al., 2010). Currently, FM is diagnosed based on subjective criteria (Wolfe et al., 2016; Wolfe et al., 2010; Wolfe et al., 1990) with no objective diagnostic tools and universally effective treatments available. This lack of definite diagnostics and treatment options may contribute to the skepticism individuals with FM frequently encounter from their social environment (Undeland & Malterud, 2007).

The pathophysiology of FM is complex and not fully understood. Research over the past decades has revealed several aberrations in FM subjects (FMS), in the central (de Tommaso et al., 2022; Julien et al., 2005; Kosek & Hansson, 1997; Lannersten & Kosek, 2010; Lee et al., 2011) and peripheral nervous system (de Tommaso et al., 2022; Evdokimov et al., 2019; Serra et al., 2014) as well as in muscle (Elvin et al., 2006; Gerdle et al., 2013; Gerdle et al., 2020). Signs of immune system activation have also been described, with elevated levels of immunomodulatory proteins both systemically (blood/serum/plasma) (Andrés-Rodríguez et al., 2020; Bäckryd et al., 2017; Uçeyler et al., 2011; Wåhlén et al., 2020) and centrally (cerebrospinal fluid) (Bäckryd et al., 2017; Kadetoff et al., 2012) and activated microglia in the brain (Albrecht et al., 2019). Intriguingly, recent data from translational studies also suggest autoimmune mechanisms in FM. Transfer of immunoglobulin G (IgG) from FMS, but not healthy controls (HC), into mice, induce FM characteristics, such as pain behavior, hypersensitivity, increased excitability of peripheral nociceptive afferents and reduced intra-epidermal nerve fiber density (Goebel et al., 2021). The IgG bound to and activated satellite glia cells (SGCs) in the dorsal root ganglia (DRG) of the mice and bound to SGCs in DRG sections from human (postmortem) donors (Goebel et al., 2021). The clinical relevance of these findings was demonstrated in a subsequent study where levels of anti-SGC IgG were determined in individual FMS. Compared to HC, two different cohorts of FMS had elevated anti-SGC IgG levels and within the FM group, these levels were associated with higher pain intensity and more severe impact of FM symptoms on daily life activities (Krock et al., 2023). Furthermore, our recent functional magnetic resonance imaging study indicated that the anti-SGC IgG predominantly affected pain transmission in the peripheral, rather than central nervous system in FMS (Fantón et al., 2023), in accordance with the translational research

(Goebel et al., 2021).

The recent findings of autoimmune components in at least a subgroup of FMS (Fantón et al., 2023; Goebel et al., 2021; Krock et al., 2023), highlight the need for an in-depth analysis of systemic inflammation in FM. Proteomic analysis of different tissues is a promising, emerging research area. Previous studies analyzed limited numbers of proteins in the serum or plasma of FMS (Uçeyler et al., 2011), or examined the broader proteome but with a small sample size (Gerdle & Ghafouri, 2020), often leading to inconsistent findings (Gerdle & Ghafouri, 2020; Uçeyler et al., 2011). Here we utilized a high-throughput method to analyze 384 proteins in serum, in a relatively large cohort composed of 93 FMS and 40 HC. Additionally, we analyzed functional protein association networks, in accordance with recommendations to use a systems biology approach (Gerdle & Ghafouri, 2020; Gomez-Varela et al., 2019), to identify potential biomarkers and diagnostic tools. The main aim was to identify a) proteins that differentiate FMS from HC and b) within the FM group, proteins that distinguish FMS with high versus low levels of anti-SGC IgG. The secondary aim was to explore the associations between serum proteins, anti-SGC IgG levels and FM symptoms.

2. Materials and methods

2.1. Study population

The study population consists of 93 FMS and 40 HC, all females (see Table 1 for characteristics). The subjects were recruited by advertisement in a local newspaper which is delivered free of charge to all households in Stockholm and surrounding municipalities. Written informed consent, in accordance with the Declaration of Helsinki was obtained from all participants, who were compensated for their invested time.

Inclusion criteria for all participants were female sex and age 20–65 years, and in addition for the FM group, fulfillment of the 1990 and 2010/2011 American College of Rheumatology (ACR) fibromyalgia criteria (Wolfe et al., 2010; Wolfe et al., 1990). Exclusion criteria for all subjects were primary causes of pain other than FM, rheumatic or autoimmune diseases, severe somatic disorders (such as neurological, cardiovascular and malignant disease), severe psychiatric disorders requiring treatment, medication with anticonvulsants or antidepressants, substance abuse, inability to refrain from hypnotics, nonsteroidal anti-inflammatory drugs, or analgesics 48 h prior to study participation,

Table 1

Subject characteristics and group differences. Abbreviations: FMS = fibromyalgia subjects, HC = healthy controls, NA = not applicable, BMI = body mass index, VAS = Visual Analogue Scale, FIQ = Fibromyalgia Impact Questionnaire, HAD=Hospital Anxiety and Depression scale, D = depression, A = anxiety, PCS=Pain Catastrophizing Scale, MFI = Multidimensional Fatigue Inventory, gen = general fatigue, PSQI=Pittsburg Sleep Quality Index, PPT=Pressure Pain Threshold, CPM=Conditioning Pain Modulation.

	FMS (n = 93) Mean (SD, min, max)	HC (n = 40) Mean (SD, min, max)	p-value
Age	46.9 (8.4, 22.0, 60.0)	48.3 (7.7, 29.0, 60.0)	0.457
BMI	26.1 (3.7, 18.4, 34.0)	22.8 (2.2, 18.6, 27.7)	<0.001
Pain duration (months)	175.9 (99.8, 12.0, 408.0)	NA	NA
FM duration (months)	125.1 (89.3, 11.0, 408.0)	NA	NA
Pain VAS average	59.0 (19.9, 18.0, 100.0)	4.2 (5.9, 0.0, 26.0)	<0.001
Pain VAS now	53.2 (21.3, 6.0, 99.0)	2.1 (3.2, 0.0, 14.0)	<0.001
Pain VAS max	76.9 (16.6, 24.0, 99.0)	11.9 (19.5, 0.0, 85.0)	<0.001
Pain VAS min	34.3 (22.3, 1.0, 97.0)	2.1 (4.6, 0.0, 21.0)	<0.001
FM Severity (FIQ)	63.0 (16.3, 13.0, 95.0)	NA	NA
Depression (HAD_D)	7.4 (4.2, 0.0, 18.0)	1.3 (1.6, 0.0, 5.0)	<0.001
Anxiety (HAD_A)	8.4 (4.7, 0.0, 21.0)	3.2 (3.1, 0.0, 11.0)	<0.001
Catastrophizing (PCS)	17.9 (10.4, 1.0, 46.0)	NA	NA
Fatigue (MFI_gen)	18.5 (2.1, 10.0, 20.0)	7.9 (3.2, 4.0, 14.0)	<0.001
Sleep (PSQI)	11.8 (3.7, 2.0, 18.0)	4.1 (2.8, 0.0, 13.0)	<0.001
Tender point count	16.5 (1.9, 11.0, 18.0)	NA	NA
Pressure pain (PPT)	152.7 (62.1, 39.0, 333.0)	322.3 (111.2, 77.0, 633.0)	<0.001
Pain modulation (CPM)	−0.05 (0.54, −0.93, 1.83)	0.22 (0.35, −0.51, 1.54)	<0.001

hypertension (>160/90 mmHg), obesity (body mass index > 35), smoking (>5 cigarettes/day), pregnancy and inability to understand and speak Swedish. Since fMRI was part of the overall project, additional exclusion criteria were previous heart or brain surgery, self-reported claustrophobia, and magnetic implants. In addition, for HCs, FM was an exclusion criterion, as well as any other pain condition with pain intensity rated as > 20/100 mm on a 100 mm visual analogue scale (VAS) where 0 corresponds to “no pain” and 100 mm to “worst imaginable pain”.

To be able to compare some of our results in FM with nociceptive pain conditions we included data from female subjects suffering from degenerative disk disease (DDD) (Rosenström et al., 2024) and osteoarthritis (OA) (Palada et al., 2020), (DDD n = 22, age: mean 44.9, SD 9.6, OA n = 17, age: mean 64.1, SD 5.4, HC n = 33, age: mean 55.2, SD 13.7) originally collected by our research group for other studies.

2.2. Procedure

The current study is part of a larger project (study plan <https://osf.io/8zqak>). Subjects were included between 2015 and 2017 (Ethical permit number: 2014/1604–31/1), and the OLINK analysis was performed in 2021. Data regarding anti-SGC IgG have previously been presented in association with FM symptom severity in a subgroup of this cohort (n = 30) (Krock et al., 2023) and in relation to cerebral pain processing in this cohort (Fantón et al., 2023).

Before the first study visit, all participants were prescreened by a nurse (via phone) regarding their health. In addition, FM patients all underwent a screening visit, including a clinical examination by a specialist in rehabilitation medicine who also ascertained that the patients fulfilled the ACR 1990 (Wolfe et al., 1990) and ACR 2010/2011 (Wolfe et al., 2010) diagnostic criteria. A blood sample was taken during the screening visit (patients) or at the start of visit one (HC). During the first study visit, all subjects completed a set of validated questionnaires, a calibration of pressure pain sensitivity and conditioning was performed (not part of this study), followed by assessment of pressure pain thresholds and conditioning pain modulation. The second visit was planned for neuroimaging (not part of this study).

2.3. Questionnaires

All participants rated their pain intensity (current and average, minimal and maximal pain during the past week) on a 100 mm Visual Analogue Scale (VAS), anchored with “no pain” and “worst imaginable pain” and completed the Hospital Anxiety and Depression Scale (HADS), the Multidimensional Fatigue Inventory (MFI) and the Pittsburg Sleep Quality Index (PSQI). In addition, FM subjects filled out the Fibromyalgia Impact Questionnaire (FIQ) and the Pain Catastrophizing Scale (PCS).

HADS consists of two subscales assessing anxiety (HAD_A) and depression (HAD_D), each subscale ranging from 0 to 21 points, with higher scores representing more severe symptoms (Bjelland et al., 2002; Zigmond & Snaith, 1983). MFI consists of 20 statements assessing five dimensions of fatigue through the subscales General Fatigue, Physical Fatigue, Mental Fatigue, Reduced Motivation and Reduced Activity (Ericsson & Mannerkorpi, 2007; Smets et al., 1995). MFI uses a 5-point Linkert scale, resulting in values between 4 and 20 points for each subscale, with higher scores indicating more severe fatigue (Ericsson & Mannerkorpi, 2007). The PSQI was developed to assess a wide variety of factors related to sleep quality during the last month. Nineteen questions generate seven sub-scores (0–3) and one total score (0–21), with higher scores representing worse sleep (Buysse et al., 1989). FIQ is a 20-item questionnaire assessing fibromyalgia symptom severity and impact on activities. A total score (0–100) is generated, where higher scores indicate a worse state (Burckhardt et al., 1991). PCS assesses three components of catastrophizing, i.e., rumination, magnification, and helplessness. It consists of 13 descriptions of feelings and thoughts, of

which occurrence in relation to the pain experience is to be rated on a 5-point scale, from 0 (not at all) to 4 (all the time). Higher scores represent higher levels of catastrophizing (Sullivan et al., 1995).

2.4. Pressure algometry

Pressure pain thresholds (PPTs) were assessed with a 1 cm² probe on a handheld algometer (Somedic Sales AB), as previously described (Kosek et al., 1993). Subjects were instructed to communicate pain sensation by pressing a button, as a response to pressure gradually increasing by approximately 50 kPa/second. PPTs were assessed bilaterally at the knee, gluteal muscle, lateral epicondyle and supraspinatus muscle, all tender point sites of the ACR 1990 fibromyalgia criteria (Wolfe et al., 1990). The average of recorded values was calculated for every individual and used for further analyses.

2.5. Conditioned pain modulation

Conditioned pain modulation (CPM) was assessed with PPTs as test stimuli and ischemic pain (Tourniquet test) as the conditioning stimulus, as previously described (Fantón et al., 2022). A baseline PPT value was assessed at the right quadriceps femoris muscle, using the pressure algometer as described above. A blood pressure cuff was inflated to 200 mmHg around the subject's upper left arm. Subjects were instructed to perform wrist extensions every 2 s holding a 1 kg weight until pain intensity ratings reached ≥ 50 mm on a 100 mm VAS scale. With the cuff inflated, PPTs were repeatedly assessed at the quadriceps femoris muscle with at least 10 s between assessments for 4 min or until the subject decided to end the procedure (end PPT value). A CPM score ((end PPT – baseline PPT)/baseline PPT) was calculated for each subject (Fantón et al., 2022). Negative scores represent pain facilitation, and positive scores represent pain inhibition.

2.6. Laboratory analyses

2.6.1. Protein assay

Proteins were measured using Olink® Explore 384 Inflammation panel in the FM cohort, and 92 Inflammation panel in the OA and DDD cohorts (Olink Proteomics AB, Uppsala, Sweden) according to the manufacturer's instructions. The analysis is based on Proximity Extension Assay (PEA), coupled with readout via next-generation sequencing (NGS). In brief, the serum samples in 96-well plates were randomized and dispensed into a 384-well incubation plate, then mixed with a solution containing oligonucleotide-conjugated antibodies, one for each of the analyzed proteins. After being incubated at 4 °C overnight, the extension solution containing PEA enzyme and reagents for polymerase chain reaction (PCR) were added. The addition of a DNA polymerase leads to the extension of the hybridized oligonucleotides, generating a unique protein identification “barcode”. Next, library preparation adds sample identification indexes and the required nucleotides for Illumina sequencing. Prior to sequencing, libraries go through a bead-based purification step, and the quality is assessed using the Agilent 2100 Bioanalyzer (Agilent Technologies, Palo Alto, CA). The raw output data undergoes quality control, normalization and conversion into NPX® values, Olink's proprietary unit of relative abundance. Assay validation data (detection limits, intra- and inter-assay precision data, predefined values, etc.) are available on the manufacturer's website (<https://www.olink.com>).

2.6.2. CD40 and CD40L ELISA

Levels of CD40 and CD40L in serum were measured by ELISA (DCCD40 and DCDL40, R&D Systems) in female patients with OA (n = 16), DDD (n = 22) and FM (n = 40) following manufacturer's instructions.

2.6.3. Assessment of anti-SGC IgG levels

The levels of anti-SGC IgG were quantified as previously described by our group (Krock et al., 2023), using an immunofluorescence assay determining the percentage of murine SGC in cell cultures binding human IgG.

2.6.3.1. Animals. Adult, (aged 12–26 weeks) female BALB/cAnNRJ mice (Janvier, France) were used for the generation of primary SGC cultures. Mice were housed in IVC GM-500 cages (Techniplast) in a dedicated temperature control animal room with a 12-hour light/dark cycle with ad libitum access to food and water. Ethical approval was received from the Stockholm North Animal Ethics Board (4945–2018).

2.6.3.2. Cell culture. The establishment of primary SGC culture was based on the protocol published elsewhere (Malin et al., 2007). Mice were deeply anesthetized with 5 % isoflurane and euthanized by decapitation. Thereafter, 35–40 dorsal root ganglia (DRG) were collected. DRG were enzymatically dissociated with papain and a collagenase/dispase blend for 30 min each, at 37 °C, in a rocking surface. After dissociation, cells were resuspended in F-12 media containing 10 % fetal bovine serum, and 1x penicillin–streptomycin. Then, a single cell suspension was achieved by gently pipetting the cells up and down to break them apart, and then filtering them through a 100 µm cell strainer. Cells were then seeded in non-coated Nunc™ Lab-Tek™ glass chamber slides. SGCs were allowed to adhere for 1.5 h, after which the supernatant, which contains neurons, was removed. Seeded cells were then washed once with fresh media and this was added again to each chamber. Cell recovery was performed overnight at 37 °C in a 5 % CO₂ incubator prior to incubation with human serum and subsequent immunocytochemistry.

2.6.3.3. Immunocytochemistry. The protocol to determine the frequency of serum IgG binding to SGCs was performed as described previously (Krock et al., 2023). Each serum sample was diluted into media at a 1:100 ratio, filtered through a 0.22 µm filter, and added to live cells for a 3-hour incubation period. This step allowed the possible recognition of cell surface and extracellular antigens by the IgG present in serum. Once the incubation time was over, cells were washed with new fresh media followed by fixation and permeabilization with 4 % formaldehyde (PFA) and 0.1 % Triton-X100 in 1X PBS, respectively. A rabbit antibody against glutamine synthetase, an established SGC marker, was then used overnight in a 1:500 dilution. Cells were washed and exposed for 1 h to AF594 anti-human IgG antibody and F488 anti-rabbit IgG antibody, in a 1:300 dilution. Later, cells were washed once more, and the nuclei were counterstained for 10 min with Hoescht. Chambers were then removed from the slides, and these were let to dry and cover-slipped with Prolong gold mounting media. Finally, a Zeiss LSM800 confocal microscope was used for cell imaging. Serum cell incubation and imaging were blindly performed to the type of sample used.

2.6.3.4. Image analysis. Analysis of IgG bound to SGC was conducted with the DRGquant pipeline (Hunt et al., 2022). In brief, single cell regions of interests (ROIs) were mapped by combining a U-net (Ronneberger et al., 2015) model trained to identify all cell soma and a stardist (Weigert & Schmidt, 2022) model trained to identify all Hoescht stained nuclei. Cell segmentation and classification were performed using FIJI (Schindelin et al., 2012) leveraging GPU acceleration via CLIJ (Haase et al., 2020). To define positive signals, such as the binding of IgG to SGC, 8xSD from the pixel intensity average for all images was used as a threshold for positive signal. Fluorescent intensity measurements were then calculated for each ROI, along with thresholded/positive area and percent positive signal. Positive cells were defined as those with IgG positive area above > 0.001 µm². All images were carefully checked to exclude non-positive cells. Finally, a python script was used to concatenate all data into one master file which was analyzed and plotted using

GraphPad Prism (version 8.4.3, GraphPad Software, San Diego, California USA).

2.7. Statistical analyses

Group differences in subject characteristics (Mann Whitney *U* test) as well as Spearman's rank correlation coefficient used for analysis of correlations between anti-SGC IgG and symptom severity were analyzed using IBM SPSS Statistics, Version 28.0.1.1 (14).

All analyses involving serum proteins were based on data after adjustment for BMI since several protein levels were shown to correlate to BMI. The BMI effect was estimated per protein based on the FM group only, using linear regression adjusting for plate and BMI. If a correlation between protein level and BMI had $p < 0.2$, the BMI effect was subtracted from the protein value (for all samples).

To investigate if protein expression can distinguish between groups (FM vs HC and FM with high (≥ 50 %) vs low (< 50 %) anti-SGC IgG) we performed partial-least square discriminant analysis (PLS-DA) modelling, implemented using the R-package ropls (Thévenot et al., 2015). The PLS-DA models were based on the BMI-corrected NPX values with at least 50 % of the values above the limit of detection (LOD). The models were evaluated in 10 5-fold cross-validations and an average accuracy was computed. Variable importance (VIP) scores were computed for all proteins and used to rank the proteins. Proteins with $VIP \geq 1.3$ were studied further. Linear regressions, adjusted for plate and BMI, were performed to study single proteins and determine if the protein was up or down regulated in FM vs HC or high (≥ 50 %) vs low (< 50 %) anti-SGC IgG. These PLS-DA and linear regression analyses were performed in R version 3.6. The corresponding PLS-DA modelling on data from female subjects with DDD (Rosenström et al., 2024), and OA (Palada et al., 2020) was performed in R version 4.5.1 using the mixomics package since these analyses were performed later and the ropls package was no longer available. The linear regressions, adjusted for plate and BMI, were performed in SPSS version 28.0.1.1 (14). Independent samples Kruskal Wallis test performed in SPSS version 28.0.0.0. was used for group comparisons between FM, OA and DDD subjects.

2.8. Protein network analyses

Proteins with $VIP \geq 1.3$ in the PLS-DA models were further analyzed using the online software tool STRING (Szklarczyk et al., 2023) (version 12.0), with default settings (interaction score with medium confidence

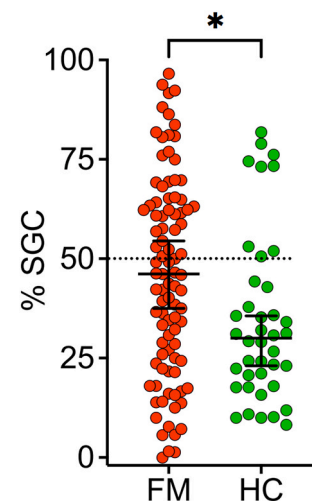


Fig. 1. The percentage of murine Satellite Glia Cells (SGC) binding human IgG. FM subjects had significantly higher anti-SGC IgG than HC ($p = 0.012$), FM: median 46.15 % (95 % CI:37.5–54.5 %), HC: median 30.02 % (95 % CI:23.1–35.6 %). *: $p < 0.05$.

Table 2

Associations between anti-SGC IgG and symptoms within the FM group. Abbreviations: VAS = Visual Analogue Scale, FIQ = Fibromyalgia Impact Questionnaire, HAD=Hospital Anxiety and Depression scale, D = depression, A = anxiety, PCS=Pain Catastrophizing Scale, MFI = Multidimensional Fatigue Inventory, gen = general fatigue, PSQI=Pittsburg Sleep Quality Index, PPT=Pressure Pain Threshold, CPM=Conditioning Pain Modulation, BMI = Body Mass Index.

	Coeff. (r)	p-value
Pain VAS average	0.421	<0.001
Pain VAS now	0.038	0.003
Pain VAS max	0.300	0.003
Pain VAS min	0.254	0.014
FM Severity (FIQ)	0.260	0.012
Depression (HAD_D)	0.233	0.024
Anxiety (HAD_A)	-0.026	0.805
Catastrophizing (PCS)	0.037	0.754
Fatigue (MFI_gen)	-0.116	0.320
Sleep (PSQI)	0.041	0.725
Tender point count	0.081	0.457
Pressure pain (PPT)	-0.120	0.303
Pain modulation (CPM)	0.080	0.501
Pain duration (months)	0.136	0.195
FM duration (months)	0.043	0.682
Age	0.082	0.43
BMI	0.050	0.641

(>0.4)), with focus on clusters based on experimentally determined functional interactions or co-expression, respectively (Szklarczyk et al., 2023). Separate analyses were performed for proteins with average values FM ≥ HC (upregulated) and proteins with FM < HC (downregulated) as well as within the FM group for proteins with higher average values in FM subjects with high (≥50 %) vs low (<50 %) anti-SGC IgG (upregulated) and proteins with lower average values in FM subjects with high vs low anti-SGC IgG (downregulated). STRING analyses were also performed with the proteins differing OA from HC and DDD from HC.

2.9. Protein categorization

Proteins were categorized based on physiological function using published references (supplementary Table 1a and b), and The Human Protein Atlas (<https://www.proteinatlas.org>).

3. Results

3.1. Subject characteristics and symptom severity

Subject characteristics and group differences are presented in Table 1. As expected, FMS score significantly higher than HC regarding all symptom related variables applicable in both groups (pain intensity (VAS), anxiety (HAD_A), depression (HAD_D), fatigue (MFI) and sleep disturbance (PSQI). FM patients also have significantly higher pain sensitivity (lower pressure pain thresholds (PPTs), as well as less pronounced descending pain inhibition (CPM). There is also a significant group difference in BMI, with FMS having higher values, which is adjusted for in the subsequent analyses. There is no significant group difference regarding age.

3.2. Anti-SGC IgG levels and their associations with symptom severity

FM subjects had significantly higher anti-SGC IgG than HC (p = 0.012), FM: median 46.15 % (95 %CI: 37.5–54.5 %), HC: median 30.02 % (95 %CI: 23.1–35.6 %) (Fig. 1). Furthermore, within the FM group, significant correlations were found between anti-SGC IgG and all pain intensity ratings (VAS), the impact of fibromyalgia (FIQ) and rated depression (HAD_D) (Table 2). No significant associations were observed between anti-SGC IgG and anxiety, pain catastrophizing, fatigue, disturbed sleep, tender points, pressure pain thresholds, conditioned pain modulation, age, pain/FM duration or BMI (Table 2). Some of these results have been presented previously in relation to cerebral pain processing (Fanton et al., 2023).

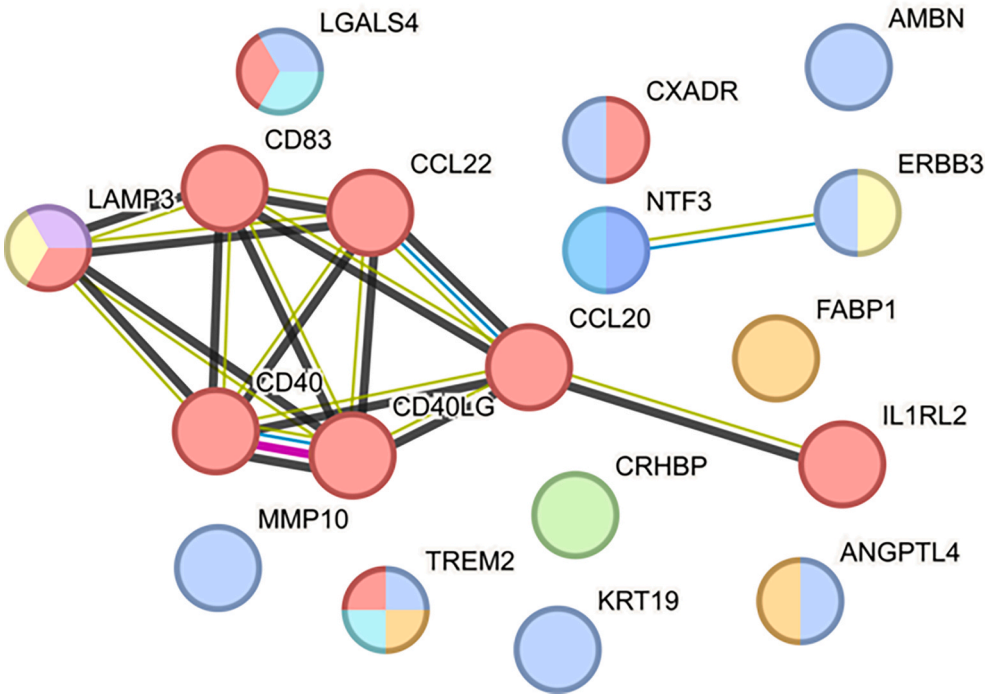


Fig. 2. Functional protein association networks (STRING) for proteins (gene names are used representing the corresponding unregulated proteins) discriminating between FMS and HC, adjusted for BMI (VIP ≥ 1.3). Proteins with higher values in FMS than HC (upregulated). Fig. 2 continued: Proteins with lower values in FMS than HC (downregulated). For protein category and protein interaction color key see Fig 5 continued.

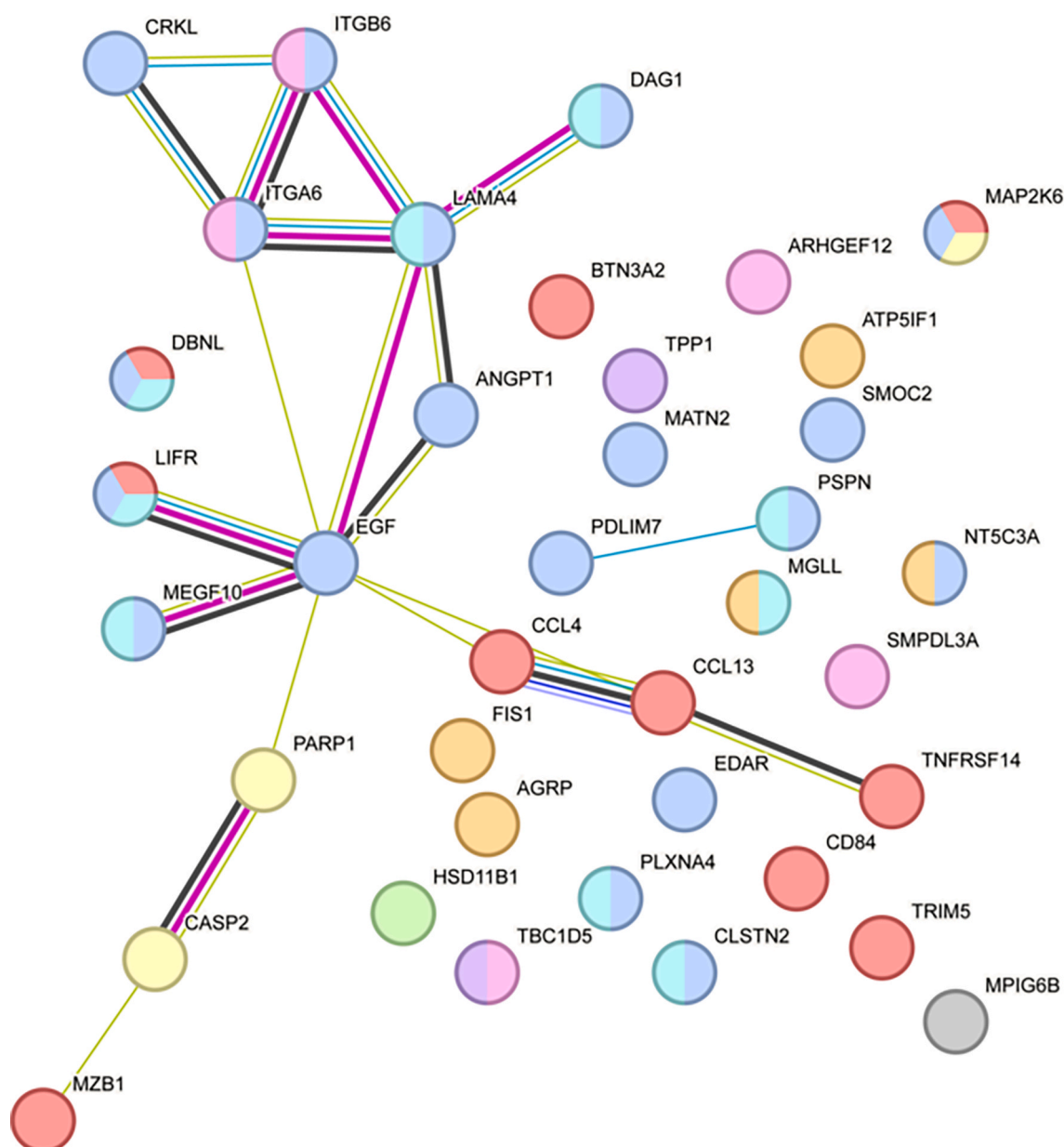


Fig. 2. (continued).

3.3. Proteins differing between FMS and HC

The PLS-DA models could distinguish between FM and HC with an average accuracy of 0.73. Fifty-six proteins had $VIP \geq 1.3$ (Fig. 2, Table 3a (including the full names of proteins) and supplementary Table 1a). To target potential pathophysiological mechanisms, we choose to dichotomize these proteins into those with higher average values in FMS than HC (upregulated, $n = 18$, Fig. 2) and those with lower average values in FMS than HC (downregulated, $n = 38$, Fig. 2 continued) and perform two separate functional protein association analyses (STRING) (Szklarczyk et al., 2023).

The upregulated proteins CD40 and CD40L have an experimentally determined functional connection (Szklarczyk et al., 2023). In addition, a cluster of co-expressed (Szklarczyk et al., 2023) proteins, all with immune system related functions were found, including CD40, CD40L, CCL20, CCL22, LAMP3, CD83 and IL1RL2 (co-expressed with CCL20). Among proteins downregulated in FMS, two clusters of proteins with experimentally determined functional connections (Szklarczyk et al., 2023) were found. One includes EGF, LIFR, MEGF10, LAMA4, DAG1,

ITGA6 and ITGB6, all involved in tissue development/maintenance/regeneration and some also involved in cell signaling. The other consists of PARP1 and CASP2 which are involved in the regulation of cell survival and apoptosis. The patterns of co-expression (Szklarczyk et al., 2023) are partially overlapping.

The two proteins with highest VIP for distinguishing FMS from HC were TREM2 (FM > HC, $p = 0.002$) and CLSTN2 (FM < HC, $p = 0.008$), none part of a functional network.

3.4. Proteins differing between FMS with high vs low anti-SGC IgG

The PLS-DA models were also run within the FM group to identify proteins distinguishing between FMS with high ($\geq 50\%$) and low ($< 50\%$) anti-SGC IgG defined as the percentage of murine SGC bound by human IgG. This model had an average accuracy of 0.51 and even though this model is not useful for prediction, VIP from the model can be used for ranking the proteins. The proteins with VIP values ≥ 1.3 ($n = 55$) are depicted in Fig. 3, Table 3b (including the full names of proteins) and supplementary Table 1b. The STRING analyses were performed

Table 3aProteins (n = 56) discriminating between FMS and HC, adjusted for BMI, in descending order of importance according to VIP (VIP $\geq$ 1.3).

Protein	Gene	VIP score	↑=upregulated in FM ↓=downregulated in FM
Triggering receptor expressed on myeloid cells 2	<i>TREM2</i>	2.3	↑
Calsyntenin 2	<i>CLSTN2</i>	2.3	↓
Persephin	<i>PSPN</i>	2.1	↓
C-C motif chemokine ligand 20	<i>CCL20</i>	1.9	↑
Laminin subunit alpha 4	<i>LAMA4</i>	1.8	↓
SPARC related modular calcium binding 2	<i>SMOC2</i>	1.7	↓
Tripeptidyl peptidase 1	<i>TPP1</i>	1.7	↓
C-C motif chemokine ligand 22	<i>CCL22</i>	1.7	↑
Integrin subunit alpha 6	<i>ITGA6</i>	1.7	↓
Ameloblastin	<i>AMBN</i>	1.7	↑
Butyrophilin subfamily 3 member A2	<i>BTN3A2</i>	1.7	↓
Galectin 4	<i>LGALS4</i>	1.6	↑
CXADR Ig-like cell adhesion molecule	<i>CXADR</i>	1.6	↑
Fatty acid binding protein 1	<i>FABP1</i>	1.6	↑
Dystroglycan 1	<i>DAG1</i>	1.6	↓
Cluster of differentiation 40 ligand (CD40L)	<i>CD40LG</i>	1.6	↑
Interleukin 1 receptor like 2	<i>IL1RL2</i>	1.6	↑
Cluster of differentiation 84	<i>CD84</i>	1.6	↓
PDZ and LIM domain 7	<i>PDLIM7</i>	1.5	↓
Plexin A4	<i>PLXNA4</i>	1.5	↓
Rho guanine nucleotide exchange factor 12	<i>ARHGEF12</i>	1.5	↓
Multiple EGF like domains 10	<i>MEGF10</i>	1.5	↓
Sphingomyelin phosphodiesterase acid like 3A	<i>SMPDL3A</i>	1.5	↓
Monoglyceride lipase	<i>MGLL</i>	1.5	↓
CRK like proto-oncogene, adaptor protein	<i>CRKL</i>	1.5	↓
TBC1 domain family member 5	<i>TBC1D5</i>	1.5	↓
Angiopoietin like 4	<i>ANGPTL4</i>	1.5	↑
Erb-b2 receptor tyrosine kinase 3	<i>ERBB3</i>	1.5	↑
Megakaryocyte and platelet inhibitory receptor G6b	<i>MPIG6B</i>	1.5	↓
Cluster of differentiation 83	<i>CD83</i>	1.5	↑
Epidermal growth factor	<i>EGF</i>	1.4	↓
Cluster of differentiation 40	<i>CD40</i>	1.4	↑
Drebrin like	<i>DBNL</i>	1.4	↓
Angiopoietin 1	<i>ANGPT1</i>	1.4	↓
Ectodysplasin A receptor	<i>EDAR</i>	1.4	↓
Keratin 19	<i>KRT19</i>	1.4	↑
Mitogen-activated protein kinase kinase 6	<i>MAP2K6</i>	1.4	↓
C-C motif chemokine ligand 4	<i>CCL4</i>	1.4	↓
Corticotropin releasing hormone binding protein	<i>CRHBP</i>	1.4	↑
Poly(ADP-ribose) polymerase 1	<i>PARP1</i>	1.4	↓
Lysosomal associated membrane protein 3	<i>LAMP3</i>	1.4	↑
Hydroxysteroid 11-beta dehydrogenase 1	<i>HSD11B1</i>	1.4	↓
TNF receptor superfamily member 14	<i>TNFRSF14</i>	1.4	↓
LIF receptor subunit alpha	<i>LIFR</i>	1.4	↓
Agouti related neuropeptide	<i>AGRP</i>	1.4	↓
ATP synthase inhibitory factor subunit 1	<i>ATP5IF1</i>	1.4	↓
Neurotrophin 3	<i>NTF3</i>	1.4	↑
Integrin subunit beta 6	<i>ITGB6</i>	1.3	↓
Caspase 2	<i>CASP2</i>	1.3	↓
Tripartite motif containing 5	<i>TRIM5</i>	1.3	↓
5'-nucleotidase, cytosolic IIIA	<i>NT5C3A</i>	1.3	↓
C-C motif chemokine ligand 13	<i>CCL13</i>	1.3	↓
Marginal zone B and B1 cell specific protein	<i>MZB1</i>	1.3	↓
Fission, mitochondrial 1	<i>FIS1</i>	1.3	↓
Matrix metalloproteinase 10	<i>MMP10</i>	1.3	↑
Matrilin 2	<i>MATN2</i>	1.3	↓

separately regarding proteins that were upregulated (Fig. 3) and downregulated (Fig. 3 continued) in FMS with high anti-SGC IgG (≥ 50 %).

Among proteins upregulated in FMS with high anti-SGC IgG, experimentally determined functional connections (Szklarczyk et al., 2023) were found in two clusters. One containing CD4, CD79b, CCL11, CCL28, CXCL12 and MMP1, all involved in immune regulation except MMP1 which is associated with tissue development/maintenance/regeneration. The second cluster consists of CKMT1A and CKMT1B, both involved in regulation of metabolism/energy homeostasis. The co-expression patterns (Szklarczyk et al., 2023) are partially overlapping.

Among proteins downregulated in FMS with high anti-SGC IgG two clusters of proteins with experimentally determined functional connections (Szklarczyk et al., 2023) were found. One cluster includes

SIGLEC1, LGALS9 and CD276, all associated with immune system regulation, and the other includes FASLG and BID, proteins regulating cell survival/apoptosis, with FASLG also involved in immune system regulation. In addition, a large cluster of fifteen co-expressed (Szklarczyk et al., 2023) proteins with mainly immune regulatory functions was found, namely IL-6, IFN γ , KYNU, IL-15, IL-18R1, IL-32, CSF3, CD276, SIGLEC1, LGALS9, SCGB1A1, AGER, BID, FASLG, and LIFR.

The two proteins with the highest VIP for distinguishing FMS with high vs low anti-SGC IgG, were IFN γ (downregulated in FMS with high anti-SGC IgG, $p = 0.04$) and GAL (upregulated in FMS with high anti-SGC IgG, $p = 0.009$).

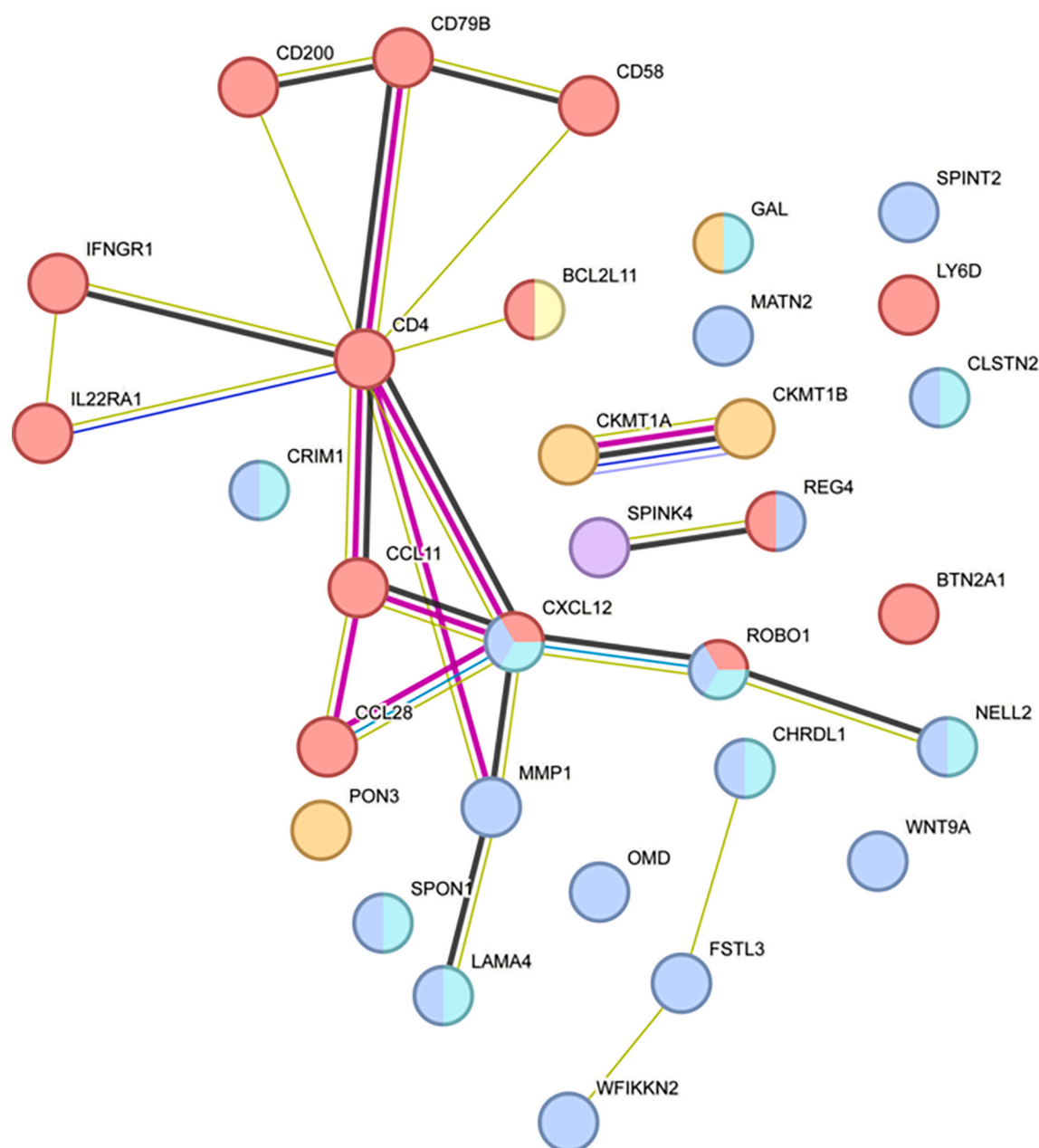


Fig. 3. Functional protein association networks (STRING) for proteins (gene names are used representing the corresponding unregulated proteins) discriminating between FMS with high ($\geq 50\%$) vs low ($< 50\%$) anti-SGC IgG, adjusted for BMI ($VIP \geq 1.3$). Proteins upregulated in FMS with high anti-SGC IgG. Fig. 3 continued: Proteins downregulated in FMS with high anti-SGC IgG. For protein category and protein interaction color key see Fig. 5 continued.

3.5. Proteins differing between FMS and HC as well as between FMS with high vs low anti-SGC IgG

Only six proteins had $VIP \geq 1.3$ both when comparing FMS with high vs low anti-SGC IgG, and when comparing FMS vs HC (CLSTN2, LAMA4, ITGA6, LIFR, ATP5IF1 and MATN2). These proteins are predominantly involved in tissue development/maintenance/regeneration (CLSTN2, LAMA4, MATN2, ITGA6 and LIFR). ITGA6 is also important for cell signaling and LIFR has immunomodulatory functions. ATP5IF1 is related to metabolism and energy homeostasis.

3.6. Proteins differing between OA and HC

The PLS-DA model on proteins analyzed in OA compared to HC

resulted in fifteen proteins with $VIP \geq 1.3$, twelve of them upregulated and three downregulated in OA (Fig. 4 and table 4). STRING analyses were performed separately regarding proteins that were upregulated and downregulated in OA. Among the upregulated proteins nine proteins, including CD40, were co-expressed (Szklarczyk et al., 2023), and two of these also showed experimentally determined functional connections (Szklarczyk et al., 2023). Eight of the co-expressed proteins are associated with immune system regulation (Table 4) (The Human Protein Atlas). Among the three proteins downregulated in OA two are co-expressed (Szklarczyk et al., 2023). Both are associated with immune system regulation, and one of them, IL17 also has a role in tissue repair (Table 4) (The Human Protein Atlas).

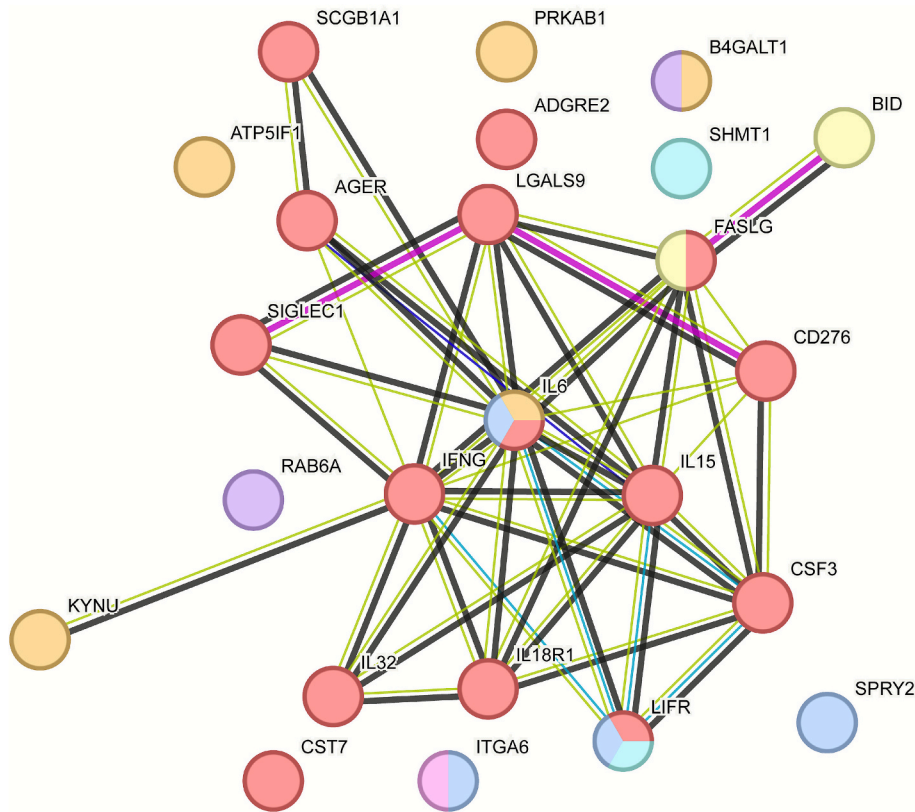


Fig. 3. (continued).

Table 3b
Proteins (n = 55) discriminating between FMS with high ($\geq 50\%$) vs low ($< 50\%$) anti-SGC IgG, adjusted for BMI, in descending order of importance according to VIP (VIP ≥ 1.3).

Protein	Gene name	VIP score	↑=upregulated in IgG high ↓=downregulated in IgG high
Interferon gamma (IFN γ)	<i>IFNG</i>	2.3	↓
Protein kinase AMP-activated non-catalytic subunit beta 1	<i>PRKAB1</i>	2.1	↓
Galanin	<i>GAL</i>	1.9	↑
Sialic acid binding Ig like lectin 1	<i>SIGLEC1</i>	1.9	↓
Matrix metalloproteinase 1	<i>MMP1</i>	1.9	↑
Kynureninase	<i>KYNU</i>	1.9	↓
C-C motif chemokine ligand 28	<i>CCL28</i>	1.8	↑
Beta-1,4-galactosyltransferase 1	<i>B4GALT1</i>	1.8	↓
Colony stimulating factor 3	<i>CSF3</i>	1.7	↓
Interleukin 22 receptor subunit alpha 1	<i>IL22RA1</i>	1.7	↑
Creatine kinase, mitochondrial 1A	<i>CKMT1A</i>	1.7	↑
Creatine kinase, mitochondrial 1B	<i>CKMT1B</i>		
Sprouty RTK signaling antagonist 2	<i>SPRY2</i>	1.6	↓
Chordin like 1	<i>CHRD1</i>	1.6	↑
Spondin 1	<i>SPON1</i>	1.6	↑
Regenerating family member 4	<i>REG4</i>	1.6	↑
Serine peptidase inhibitor, Kunitz type 2	<i>SPINT2</i>	1.6	↑
Adhesion G protein-coupled receptor E2	<i>ADGRE2</i>	1.5	↓
BCL2 like 11	<i>BCL2L11</i>	1.5	↑
C-X-C motif chemokine ligand 12	<i>CXCL12</i>	1.5	↑
Cluster of differentiation 4	<i>CD4</i>	1.5	↑
Interferon gamma receptor 1	<i>IFNGR1</i>	1.5	↑
Calsyntenin 2	<i>CLSTN2</i>	1.5	↑
Fas ligand	<i>FASLG</i>	1.5	↓
Follistatin like 3	<i>FSTL3</i>	1.5	↑
Butyrophilin subfamily 2 member A1	<i>BTN2A1</i>	1.5	↑
BH3 interacting domain death agonist	<i>BID</i>	1.5	↓
Interleukin 18 receptor 1	<i>IL18R1</i>	1.5	↓
Cluster of differentiation 58	<i>CD58</i>	1.5	↑
Matrilin 2	<i>MATN2</i>	1.5	↑
Cluster of differentiation 79b	<i>CD79B</i>	1.5	↑
Serine hydroxymethyltransferase 1	<i>SHMT1</i>	1.4	↓
Cysteine rich transmembrane BMP regulator 1	<i>CRIM1</i>	1.4	↑

(continued on next page)

Table 3b (continued)

Protein	Gene name	VIP score	↑=upregulated in IgG high ↓=downregulated in IgG high
Osteomodulin	OMD	1.4	↑
Neural EGFL like 2	NELL2	1.4	↑
Cystatin F	CST7	1.4	↓
Galectin 9	LGALS9	1.4	↓
Interleukin 6	IL6	1.4	↓
Advanced glycosylation end-product specific receptor	AGER	1.4	↓
Secretoglobin family 1A member 1	SCGB1A1	1.4	↓
ATP synthase inhibitory factor subunit 1	ATP5IF1	1.4	↓
WAP, follistatin/kazal, immunoglobulin, kunitz and netrin domain containing 2	WFIKKN2	1.4	↑
Interleukin 15	IL15	1.4	↓
C-C motif chemokine ligand 11	CCL11	1.4	↑
Laminin subunit alpha 4	LAMA4	1.4	↑
Cluster of differentiation 200	CD200	1.4	↑
Wnt family member 9A	WNT9A	1.4	↑
Lymphocyte antigen 6 family member D	LY6D	1.4	↑
LIF receptor subunit alpha	LIFR	1.4	↓
Paraoxonase 3	PON3	1.3	↑
Cluster of differentiation 276	CD276	1.3	↓
Interleukin 32	IL32	1.3	↓
RAB6A, member RAS oncogene family	RAB6A	1.3	↓
Integrin subunit alpha 6	ITGA6	1.3	↓
Roundabout guidance receptor 1	ROBO1	1.3	↑
Serine peptidase inhibitor Kazal type 4	SPINK4	1.3	↑

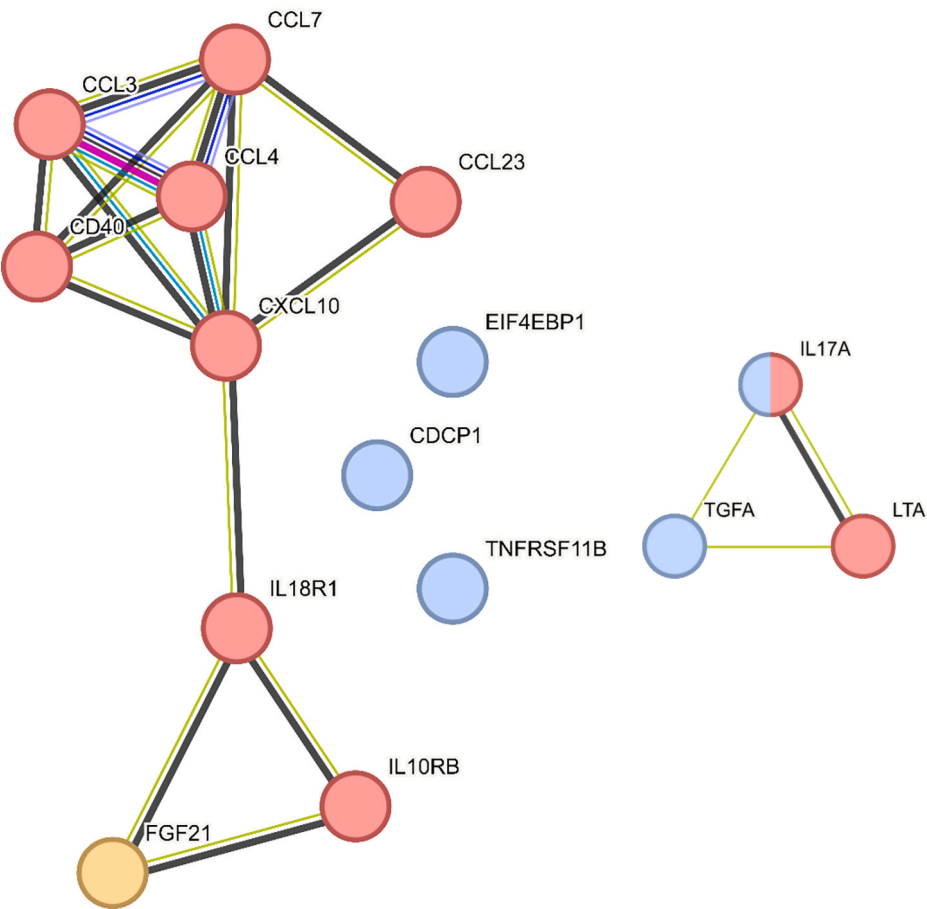


Fig. 4. Functional protein association networks (STRING) for proteins (genenames are used representing the corresponding unregulated proteins) discriminating between OA and HC, adjusted for BMI (VIP ≥ 1.3). Left: Proteins upregulated in OA. Right: Proteins downregulated in OA. For protein category and protein interaction color key see Fig. 5 continued.

3.7. Proteins differing between DDD and HC

The PLS-DA model on proteins analyzed in DDD compared to HC resulted in sixteen proteins with VIP ≥ 1.3, all of them downregulated in

DDD (Fig. 5 and table 5). The STRING analysis resulted in a cluster of fifteen co-expressed proteins (Szkłarczyk et al., 2023), including CD40. Eight of them also have experimentally determined functional connections (Szkłarczyk et al., 2023). Eleven of the fifteen proteins included in

Table 4Proteins (n = 15) discriminating between OA and HC adjusted for BMI, in descending order of importance according to VIP (VIP $\geq$ 1.3).

Protein abbreviation/ gene name	VIP score	↑=upregulated in OA ↓=downregulated in OA	Functional category	Function
IL18R1	3.0	↑	Immune system functions/Immune response modulator	Cytokine receptor, for IL-18 (proinflammatory), may also bind IL-37 (anti-inflammatory).
FGF21	2.5	↑	Metabolism and energy homeostasis	Growth factor with main function to stimulate the uptake of glucose in adipose tissue, regulates glucose homeostasis.
CDCP1	2.0	↑	Tissue development, maintenance and regeneration	May be involved in cell adhesion and proliferation and cell matrix association.
TGF α	1.9	↓	Tissue development, maintenance and regeneration	Growth factor.
CD40	1.9	↑	Immune system functions/Immune response modulator	Immune receptor involved in cellular and humoral adaptive immunity.
4EBP1/EIF4EBP1	1.8	↑	Tissue development, maintenance and regeneration	Translation repression, protein synthesis inhibition
TNFB = LTA	1.8	↓	Immune system functions/Immune response modulator	Mediates a variety of inflammatory, immunostimulatory, and antiviral responses.
CXCL10	1.6	↑	Immune system functions/Immune response modulator	Proinflammatory cytokine involved in several immunological processes.
CCL3	1.6	↑	Immune system functions/Immune response modulator	Cytokine involved in chemotaxis and inflammation.
CCL23	1.5	↑	Immune system functions/Immune response modulator	Cytokine involved in chemotaxis and inflammation.
MCP3 = CCL7	1.5	↑	Immune system functions/Immune response modulator	Cytokine involved in chemotaxis and inflammation.
IL17A	1.5	↓	Immune system functions/Immune response modulator	Cytokine with important functions in immune responses and has a role in tissue repair.
IL10RB	1.4	↑	Tissue development, maintenance and regeneration	Cytokine receptor. Antiviral defense.
CCL4	1.4	↑	Immune system functions/Immune response modulator	Cytokine involved in chemokinetic functions and inflammation.
OPG = TNFRSF11B	1.4	↑	Tissue development, maintenance and regeneration	Inhibits activation of osteoclasts and promotes osteoclast apoptosis.

the cluster based on co-expression are associated with immune system regulation, two of them also have tissue related functions (Table 5) (The Human Protein Atlas), (Böttinger et al., 1996). Of the remaining proteins two have tissue related functions and one has a role in fibrinolysis (Table 5) (The Human Protein Atlas).

3.8. Comparisons across patient groups

There were statistically significant differences in CD40 levels between FMS and female patients suffering from painful knee OA ($p = 0.027$) or from DDD ($p < 0.001$). A significant difference was also seen between the OA and DDD groups ($p < 0.001$). FMS CD40 levels were in between those of the OA group (highest) and the DDD group (lowest) (Supplementary Fig. 1). We found no statistically significant difference in CD40L between FMS and female patients suffering from painful knee OA or DDD (supplementary Fig. 2).

3.9. Associations between serum proteins and FM symptoms

To assess the potential clinical relevance of the serum proteins, we analyzed associations between proteins and symptom severity in the FMS. The correlation coefficients are presented for all significant correlations ($p < 0.05$) between symptoms and proteins distinguishing between FMS/HC (Fig. 6) or FMS with high vs low anti-SGC IgG (Fig. 6 continued), respectively. Regarding proteins differentiating between FMS/HC, we found one upregulated protein in FMS which was associated with more severe symptoms (higher FIQ ratings), namely CD40. Among the downregulated proteins in FMS, two were associated with more severe symptoms, that is ANGPT1 (higher FIQ ratings) and CLSTN2 (higher VAS pain ratings), and two with lower pain intensities, namely ATP5IF1 and CCL13.

Five proteins upregulated in FMS with high vs low anti-SGC IgG were

associated with symptom severity. Four of these were associated with increased symptom severity namely, SPON1 (higher FIQ ratings), CLSTN2 (higher VAS ratings), LY6D and CD79B (increased pressure pain sensitivity), while only one, NELL2 was inversely correlated with pain. Out of the proteins downregulated in FMS with high anti-SGC IgG, nine were associated with symptom severity. Seven of these inversely correlated with pain (IFN γ , CST7, ATP5IF1, B4GALT1, SIGLEC, IL-6 and CSF3). In addition, CST7 was also negatively associated with FM severity (FIQ ratings) while CSF3 was also associated with reduced pressure pain sensitivity. The remaining two were associated with more pain (IL-15) and increased pressure pain sensitivity (SCGB1A1).

ATP5IF1 was downregulated in FMS compared to HC as well as in FMS with high vs low levels of anti-SGC IgG and was negatively correlated to pain intensity.

4. Discussion

4.1. Main findings

Among proteins differentiating between FMS and HC, a cluster of seven co-expressed (Szklaarczyk et al., 2023) proteins upregulated in FMS was found, all related to immune system regulation. This cluster included CD40 and CD40L, having central roles in humoral immune response (Elgueta et al., 2009). Moreover, stressing the clinical relevance of this finding, CD40 levels were associated with higher impact of FM symptoms. We compared these results with other pain conditions (OA and DDD) characterized by mainly nociceptive pain mechanisms. OA patients had a cluster of eight up-regulated inflammatory proteins, including CD40, however, none of the other proteins in this cluster were found to be upregulated in FM patients, in fact, CCL4, upregulated in OA was downregulated in FM. In addition, DDD patients did not have any upregulated proteins, but a large cluster of down-regulated

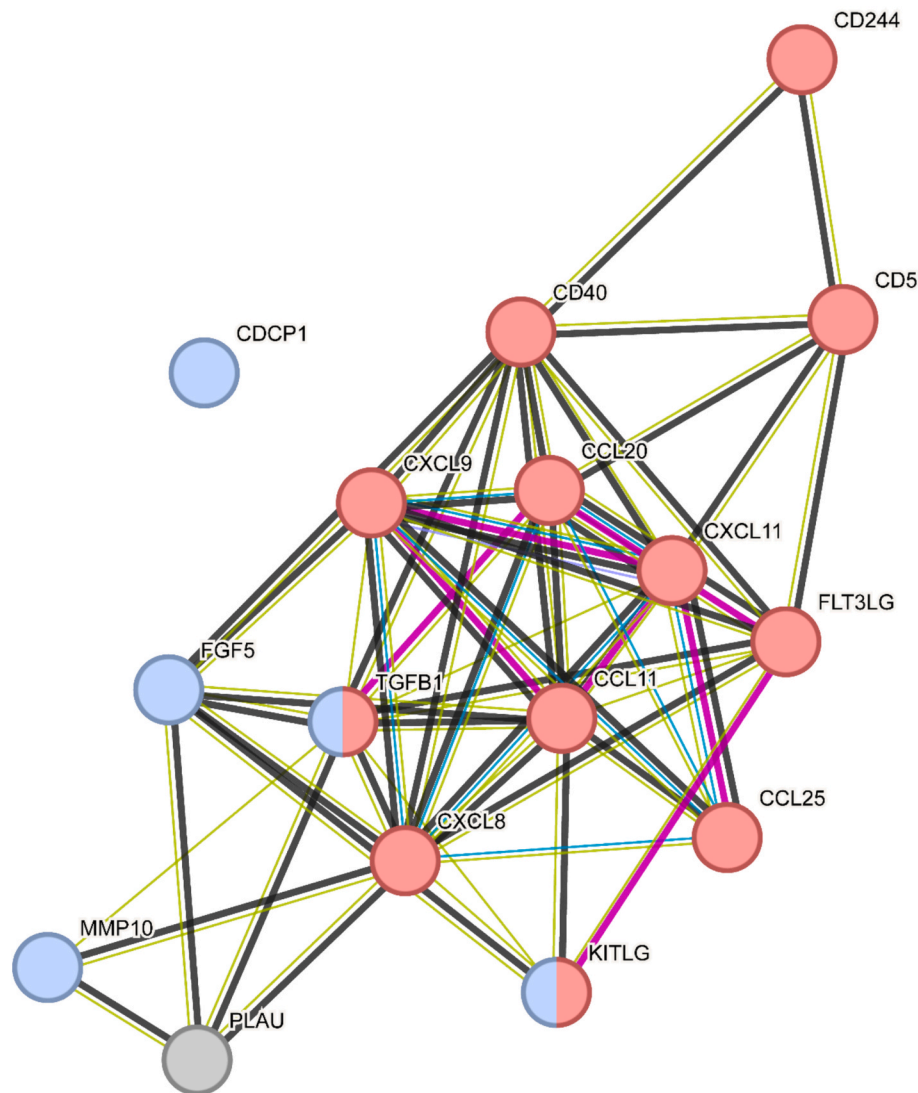


Fig. 5. Functional protein association networks (STRING) for proteins (gene names are used representing the corresponding unregulated proteins) discriminating between DDD and HC, adjusted for BMI ($VIP \geq 1.3$). Proteins downregulated in DDD (there are no upregulated proteins). Note that LAPTGFb (Latency Associated Peptide of TGFbeta) is not available to add to STRING analysis why TGFB1 was entered instead. Fig. 5 continued: Protein category and protein interaction color key.

inflammatory proteins was found, including CD40, CCL20 and MMP10, all upregulated in FM. Additional ELISA analyses showed that CD40 levels in FMS were in between those observed in female OA and DDD subjects. Interestingly, the main cluster of proteins downregulated in FMS consisted of proteins related to tissue development, maintenance and regeneration, without significant associations with symptom severity. This finding was unique for the FM cohort. In conclusion, our results indicate disease specific patterns of up- or downregulated proteins, not related to nociceptive pain. It remains to be determined if the protein patterns seen in our FM patients are specific for FM, or seen across nociplastic pain conditions.

Regarding proteins differentiating between FMS with high vs low anti-SGC IgG, a cluster, based on experimentally determined functional connections (Szklarczyk et al., 2023), of six upregulated proteins in FMS with high anti-SGC IgG was found. Five of these were related to immune system regulation, including CD4, a co-receptor on T-helper cells involved in activation of various immune cells, including B-cells and macrophages (Luckheeram et al., 2012) and CD79b, a protein necessary for B cell activation upon antigen recognition (Huang et al., 2011). Furthermore, some of these upregulated proteins were associated with symptom severity, specifically pain intensity (CLSTN2), impact of FM

(SPON1) and pressure pain sensitivity (CD79b, LY6D). Interestingly, a large cluster of proteins with known immunoregulatory functions (such as $IFN\gamma$ and IL-6) were downregulated in FMS with high anti-SGC IgG. Many of these proteins were inversely correlated to pain intensity, suggesting protective effects. We found a very limited overlap between proteins differentiating between FM patients and HC and those related to the levels of anti-SGC IgG within the FM group, which is in accordance with the view of FM as a heterogenous condition regarding pathophysiological mechanisms.

As reported previously (Fanton et al., 2023; Krock et al., 2023), FMS had significantly higher levels of anti-SGC IgG than HC. Furthermore, the strongest correlations between anti-SGC IgG were with pain intensity, while no significant correlations were seen with measures of pressure pain sensitivity, the function of descending pain inhibition (CPM) nor with other typical FM symptoms such as fatigue and disturbed sleep. These results indicate that anti-SGC IgG are primarily related to the severity of spontaneous ongoing pain in FM, which is in agreement with the results from our previous neuroimaging study of this cohort (Fanton et al., 2023). Overall, our data support the involvement of autoimmune mechanisms in a subgroup of FMS.

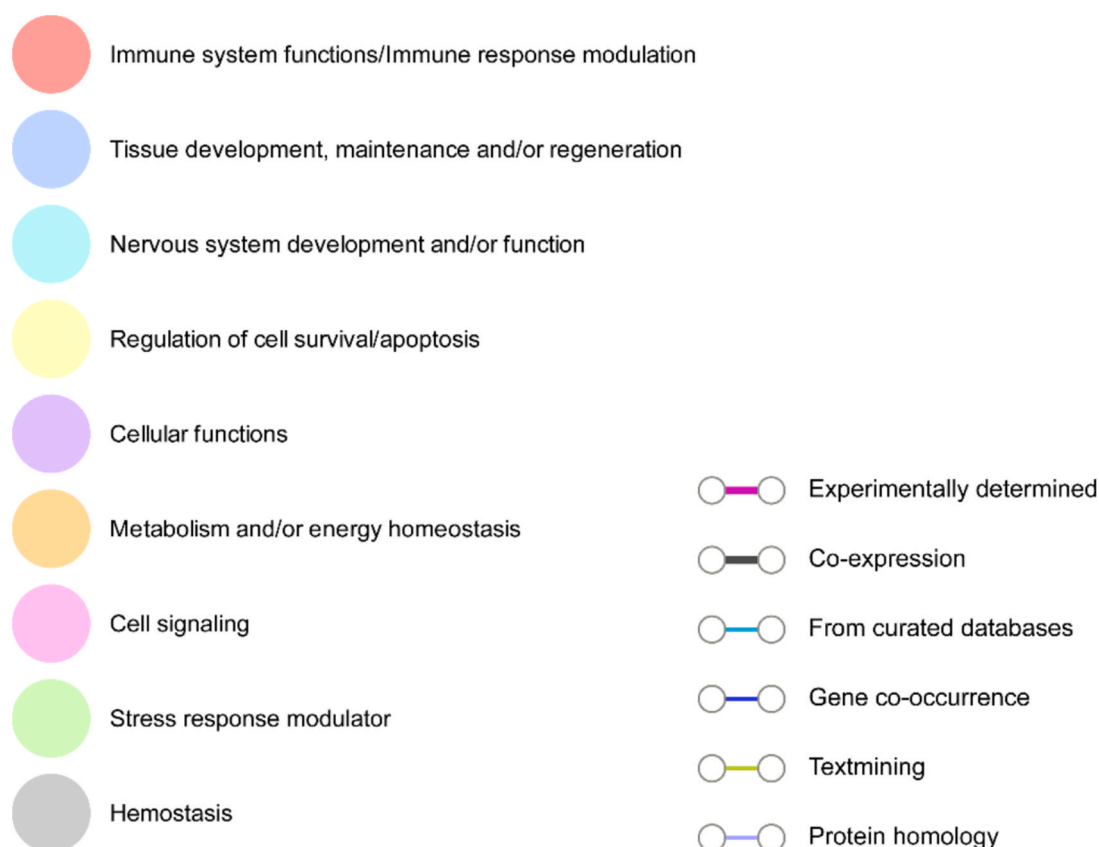


Fig. 5. (continued).

Table 5

Proteins (n = 16) discriminating between DDD and HC adjusted for BMI, in descending order of importance according to VIP (VIP ≥ 1.3). Note that LAPTGFb (Latency Associated Peptide of TGFbeta) is not available to add to STRING analysis why TGFB1 was entered instead.

Protein abbreviation/gene name	VIP score	↑=upregulated in DDD ↓=downregulated in DDD	Functional category	Function
uPA=PLAU	1.8	↓	Hemostasis/Fibrinolysis	Fibrinolysis is the primary function of uPA.
IL8=CXCL8	1.7	↓	Immune system functions/Immune response modulator	Mediates inflammatory response by attracting neutrophils, basophils, and T-cells.
CD5	1.7	↓	Immune system functions/Immune response modulator	Regulation of TCR and BCR signaling, thymocyte selection, T-cell effector differentiation and immune tolerance.
Flt3L = FLT3LG	1.6	↓	Immune system functions/Immune response modulator	Controls development of dendritic cells.
FGF5	1.6	↓	Tissue development, maintenance and regeneration	Regulation of cell proliferation and cell differentiation. Required for normal regulation of the hair growth cycle.
CXCL9	1.5	↓	Immune system functions/Immune response modulator	Affects the growth, movement, or activation state of cells that participate in immune and inflammatory response. Chemotactic for activated T-cells.
LAPTGFb = Latency Associated Peptide of TGFbeta Not available to add to STRING analysis why TGFB1 was entered instead	1.5	↓	Tissue development, maintenance and regeneration and Immune system functions/Immune response modulator	Put TGFbeta in a latent state. TGFbeta regulates cell proliferation, differentiation and growth of various cell types with various functions including tissue development and immune functions and can modulate expression and activation of other growth factors. (Böttinger et al., 1996)
CDCP1	1.4	↓	Tissue development, maintenance and regeneration	Cell adhesion and cell matrix association.
CCL25	1.4	↓	Immune system functions/Immune response modulator	Chemotactic activity for dendritic cells, thymocytes, and activated macrophages but is inactive on peripheral blood lymphocytes and neutrophils.
MMP10	1.4	↓	Tissue development, maintenance and regeneration	Matrix metalloproteinase, involved in ECM breakdown.
CD40	1.4	↓	Immune system functions/Immune response modulator	Immune receptor involved in cellular and humoral adaptive immunity.
CD244	1.4	↓	Immune system functions/Immune response modulator	Modulation of NK-cell cytolytic activity.

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Table 5 (continued)

Protein abbreviation/gene name	VIP score	↑=upregulated in DDD ↓=downregulated in DDD	Functional category	Function
SCF = KITLG	1.4	↓	Tissue development, maintenance and regeneration and Nervous system development Immune system functions/Immune response modulator	Hematopoiesis, germ cell and neural cell development, mast cell development, migration and function.
CXCL11	1.4	↓	Immune system functions/Immune response modulator	Induces a chemotactic response in activated T-cells.
CCL11 = Eotaxin	1.4	↓	Immune system functions/Immune response modulator	Eosinophil specific chemoattractant.
CCL20	1.4	↓	Immune system functions/Immune response modulator	Cytokine, chemokine. Relevance in autoimmunity.

4.2. Proteins distinguishing between FMS and HC

The STRING interaction-analysis of proteins upregulated in FMS, resulted in a cluster of seven co-expressed proteins (Szklarczyk et al., 2023) (CD40, CD40L, CCL20, CCL22, LAMP3, CD83, IL1RL2) related to the function and response modulation of the immune system (Comerford et al., 2010; de Saint-Vis et al., 1998; Elgueta et al., 2009; Grosche et al., 2020; Rapp et al., 2019; Vigne et al., 2011) (Fig. 2). CD40 and CD40L are both transmembrane proteins belonging to the TNF receptor/ligand family linked to many inflammatory and autoimmune diseases (Kawabe et al., 2011). Examples of CD40/CD40L mediated effects are T cell-dependent B cell proliferation, promotion of antigen presenting cells activation, immunoglobulin (Ig) isotype switching, modification of Ig to enhance affinity to antigen, and formation of memory B-cells, and CD40L gene mutation in humans results in reduced serum levels of IgG (Elgueta et al., 2009). However, as our analyte (serum) is cell-free, we must consider also the role of soluble CD40/CD40L. While the soluble form of CD40L has been reported to have biological activities similar to the transmembrane form (Kawabe et al., 2011), the soluble form of CD40, released by B cells binds CD40L in vitro (Schwabe et al., 1999), but, to our knowledge its functions in vivo are not known (Schwabe et al., 1999). In accordance with our results, previous studies using PEA technology (OLINK 92 inflammatory panel) reported elevated CD40 levels in FM (plasma) (Bäckryd et al., 2017), chronic widespread pain (plasma) (Hysing et al., 2019), diabetic polyneuropathy (serum) (Bäckryd et al., 2022) and knee osteoarthritis (serum) (Palada et al., 2020), while patients with DDD and lumbar disc herniation, respectively, had lower levels (serum) (Rosenström et al., 2024), compared to HC. In alignment with this, our ELISA analysis showed that FMS present with CD40 levels in between those observed in female subjects of Paladás OA cohort (Palada et al., 2020) and Rosenströms DDD cohort (Rosenström et al., 2024) (supplementary figure 1). CD40 was the only upregulated protein associated with more severe FM symptoms in our study. Previously, CD40 levels were reported to correlate with the intensity of neuropathic (Bäckryd et al., 2022), but not nociceptive pain (Palada et al., 2020; Rosenström et al., 2024), suggesting that CD40 may be especially relevant for the pathophysiology of nociplastic and neuropathic pain conditions. Another noteworthy member of the cluster of upregulated proteins is CCL20. Together with its receptor CCR6, CCL20 is described to be relevant in autoimmunity and as a promising therapeutic target in patients with inflammatory bowel disease, psoriasis, rheumatoid arthritis (RA), and multiple sclerosis (Meitei et al., 2021). However, it did not correlate to symptoms in our study.

Regarding downregulated proteins in FMS, a large cluster of nine proteins, with experimentally determined functional interactions and/or co-expressions (Szklarczyk et al., 2023), all associated with tissue (including nervous system) development, maintenance and regeneration (Barresi & Campbell, 2006; Beck et al., 1990; Birge et al., 2009; Chung et al., 2013; Davis & Pennypacker, 2018; Meecham & Marshall, 2020; Thurston, 2002; Zeng & Harris, 2014) was found (Fig. 2

continued). No tissue-related protein cluster similar to this was found in OA or DDD, suggesting it may be characteristic of FM. However, none of these proteins correlated with symptom severity. Finally, three co-expressed, downregulated proteins (CCL4, CCL13 and TNFRSF14) were seen, all three with immunomodulatory functions (Mendez-Enriquez & García-Zepeda, 2013; Mukaida et al., 2020; Steinberg et al., 2011). One of them, CCL13, a known regulator of immune cell migration (Mendez-Enriquez & García-Zepeda, 2013), was negatively associated with pain intensity, suggesting a protective/analgesic role.

The upregulation of CCL20, CD40 and MMP10 in FM, and of CD40 also in OA, contrasted with their downregulation in DDD, is interesting. Especially in the context of predominantly upregulated immune modulating proteins in OA and FM, predominantly downregulated immune modulating proteins in DDD and the dominance of tissue related proteins among the ones downregulated in FM. This might reflect assumed mechanistic differences. With FM being characterized by low grade immune system activation and possibly imbalance in tissue homeostasis, DDD by structural tissue injury with peripheral inflammation where the inflammatory response might be attenuated in a chronic phase, and OA by a more active inflammatory state, of which clinical signs are often observable. MMP10, which is a tissue related protein, being upregulated in FM would not contrast with the assumption of imbalanced tissue homeostasis in FM. MMP10 is a matrix metalloproteinase, involved in extra cellular matrix protein breakdown (Batra et al., 2012).

The two proteins with the highest VIP scores distinguishing FMS from HC were TREM2 (upregulated in FMS, $p = 0.002$) and CLSTN2 (downregulated in FMS, $p = 0.008$). These did not form part of the functional networks described above. TREM2 promotes activation and survival of macrophages and microglia, has complex immunomodulatory and possibly also neuroprotective effects (Deczkowska et al., 2020) and can induce mechanical allodynia in animals (Kobayashi et al., 2016). No significant correlations between TREM2 and FM symptoms were found. CLSTN2 is one of three similar membrane proteins, calyntenin 1–3, which are predominantly found in the postsynaptic membrane of neurons (Hintsch et al., 2002). The calyntenins are shown to be involved in neuroplasticity by enhancing formation and function of (inhibitory) synapses (Um et al., 2014). Accordingly, it could be of relevance for nociplastic mechanisms, which is also supported by the positive correlation between CLSTN2 levels and pain intensity in our FMS.

In summary, we found evidence suggesting the role of immunomodulatory/inflammatory proteins in the pathophysiology of FM, tallying previous results of altered concentrations of proteins involved in immunity, inflammation and metabolism in FMS and their association with FM symptoms (Wählén et al., 2020). In addition, our results indicate a disease specific pattern of protein dysregulation in FMS, with downregulation of several tissue related proteins, not seen in patients suffering from pain due to OA or DDD.

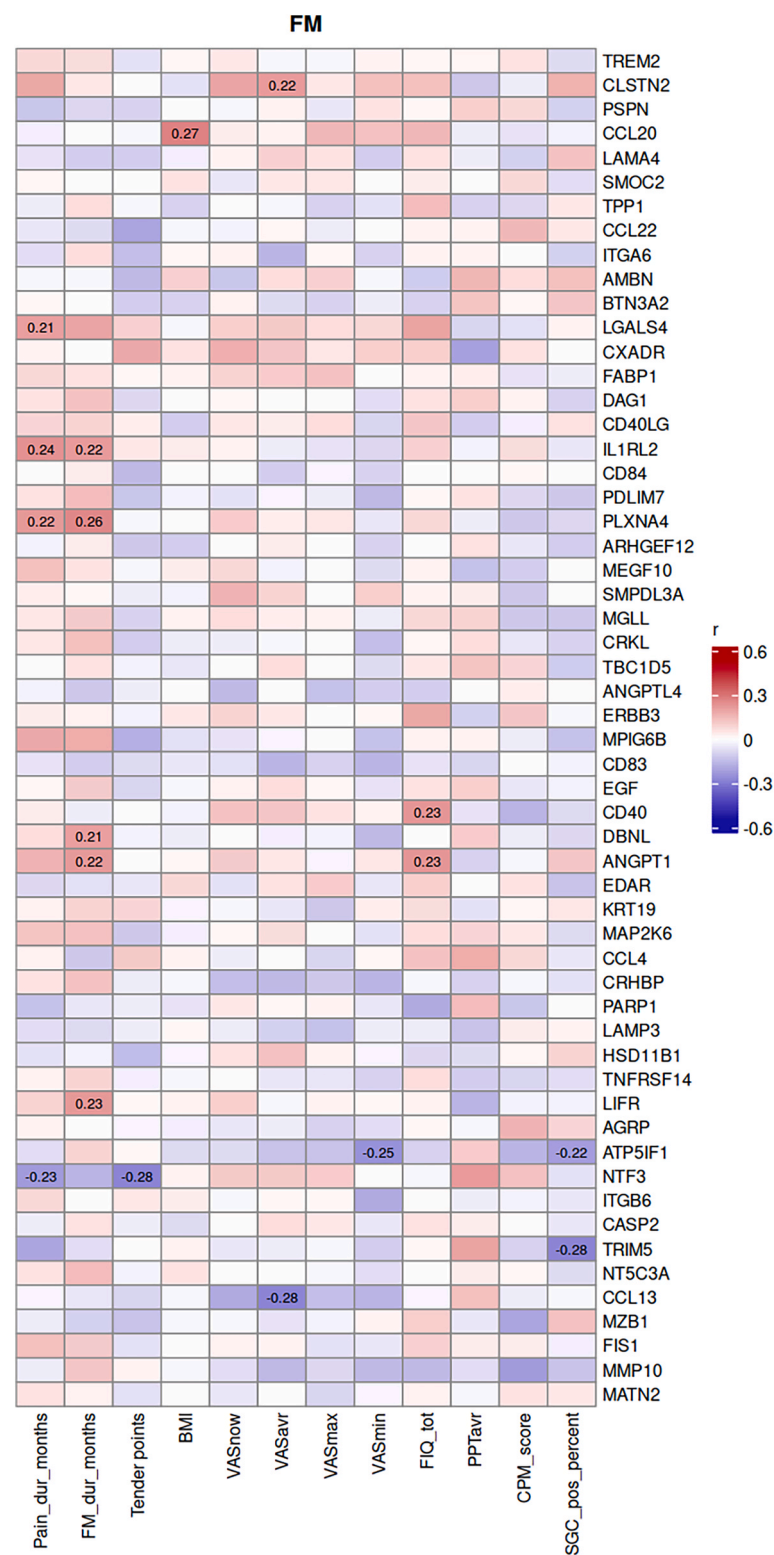


Fig. 6. Proteins discriminating FMS from HC (Fig. 6) and proteins discriminating FMS with high levels from FMS with low levels of anti-SGC IgG (Fig. 6 continued), in descending order of importance according to VIP (VIP ≥ 1.3), adjusted for BMI. The associations according to Spearman's rank correlation test in the FM group between the protein levels and the duration of pain/FM, number of tender points, body mass index (BMI), visual analogue scale (VAS) ratings of current pain intensity at investigation (VASnow), average weekly pain intensity (VASavr), maximal weekly pain intensity (VASmax), minimal weekly pain intensity (VASmin), impact of fibromyalgia (fibromyalgia impact questionnaire total score, FIQ_tot), average pressure pain threshold (PPTavr), descending pain inhibition (conditioned pain modulation, CPM_score), % murine satellite glia cells binding human IgG (SGC_pos_percent).

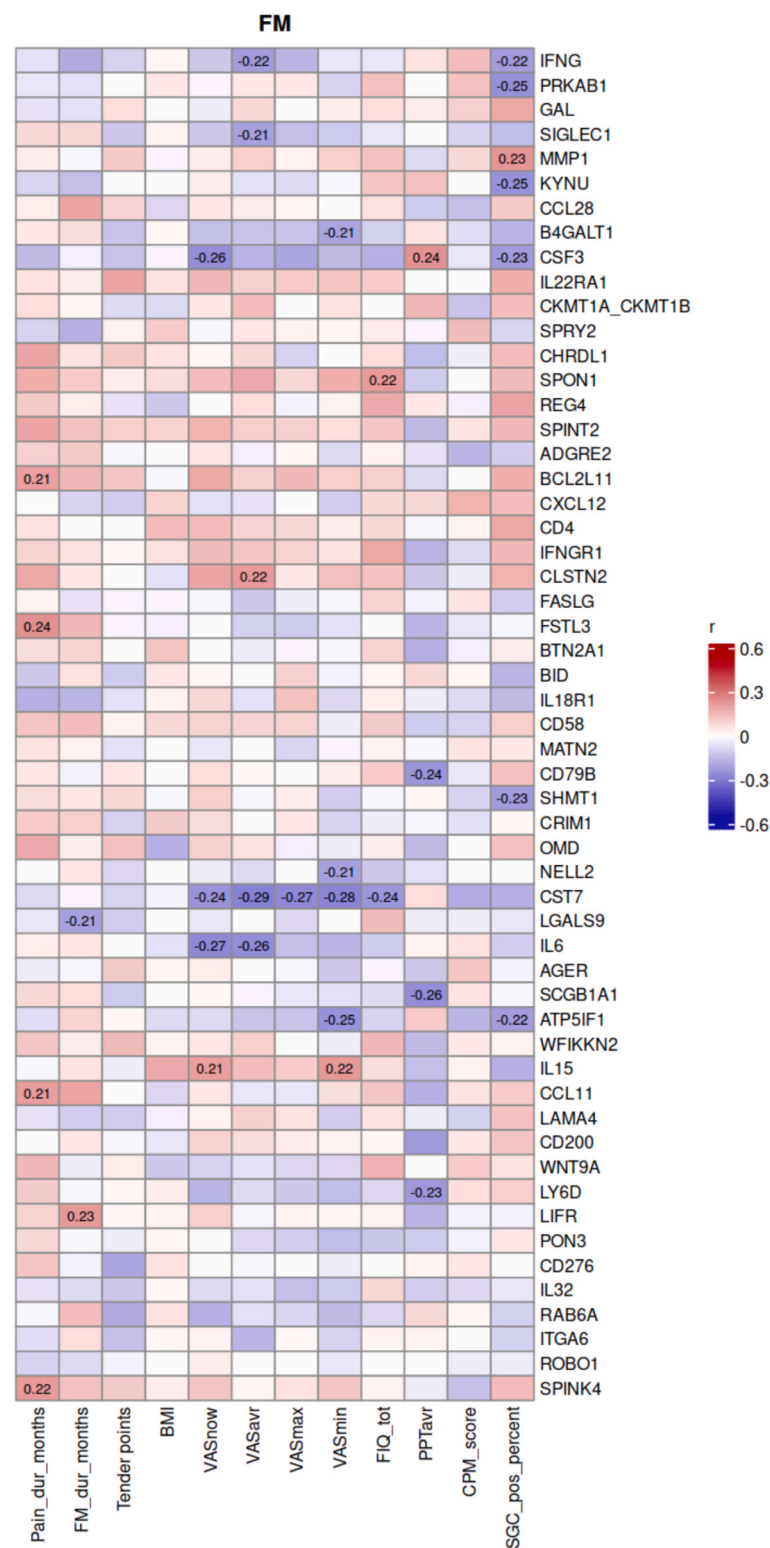


Fig. 6. (continued).

4.3. Proteins distinguishing between FMS with high vs low anti-SGC IgG

Regarding the proteins distinguishing between FMS with high vs low anti-SGC IgG the STRING protein-interaction analysis, of proteins upregulated in FMS with high anti-SGC IgG (Fig. 3), revealed a cluster with experimentally determined functional interactions (Szklaarczyk et al., 2023). Five are associated with immune functions (CD4, CD79b,

eotaxin (CCL11), CXCL12, CCL28) (Janssens et al., 2018; Luckheeram et al., 2012; Mohan et al., 2017; Rankin et al., 2000; Tkachenko et al., 2023). Co-expressed (Szklaarczyk et al., 2023) proteins partially overlap. CD4 and CD79b are of particular interest. CD4 is a co-receptor for MCH class II-restricted T lymphocyte activation, involved in the communication between T cells and antigen presenting cells (i.e. B-cells, macrophages and dendritic cells) (Künzli & Masopust, 2023). CD79b, forms

part of the signal transduction region (Ig- β) associated with the B-cell antigen receptor, and modulates the B-cells response to antigens by regulating B-cell activation, growth and differentiation (Huang et al., 2011). Therefore, both CD4 and CD79b are of vital importance for regulation of adaptive immune responses and antibody production. In our study, a positive association was found between CD79b and pressure pain sensitivity in FMS.

GAL was the protein with the highest VIP-value among the upregulated proteins, but not part of any clusters. GAL is a neuroendocrine peptide found in the central and peripheral nervous system, known to regulate various physiological processes (Lang et al., 2007) including immunoregulation, nervous system development and regeneration and nociception (Ramsbacher et al., 2019). Interestingly, GAL has been reported to affect IFN γ production and secretion by natural killer (NK)-cells (Koller et al., 2017) and is able to inhibit IFN γ stimulated production of proinflammatory cytokines in monocytes (Ramsbacher et al., 2019). Thus hypothetically, GAL could be functionally associated with the proteins exhibiting reduced levels in FMS with high anti-SGC IgG, discussed below, particularly IFN γ .

Among the proteins downregulated in FMS with high anti-SGC IgG (Fig. 3 continued), several show inverse correlations to symptom severity. The cluster of 15 co-expressed proteins is dominated by proteins with immune system related functions. It includes IFN γ , which is also the protein with the highest VIP score differentiating FMS with high vs low anti-SGC IgG. IFN γ is a functionally pleiotropic cytokine, that can modulate expression of numerous genes. It is important in both innate and adaptive immunity (Lees, 2015) and is the principal T helper cell type 1 associated cytokine. It induces production of proinflammatory cytokines, increases expression of toll like receptors and MHC class I and II, effectively activates macrophages, and microglia, and promotes phagocytosis (Kann et al., 2022; Lees, 2015; Mosmann & Coffman, 1989). IFN γ has been described to increase C-fiber activity and induce hyperalgesia in rats (Tan et al., 2021). The role of IFN γ in autoimmune disease has been investigated with ambiguous results. In neuro-inflammatory autoimmune disease, as well as in RA, IFN γ seems to be able to have both aggravating and ameliorating effects depending on the timing and context (Gauthier & Chen, 2022; Lees, 2015). In our study, FMS with high anti-SGC IgG had lower levels of IFN γ ($p = 0.04$) and a negative correlation was seen between IFN γ and pain intensity, suggesting that IFN γ may have a protective role in FM.

Three additional proteins, co-expressed with IFN γ , were also negatively associated with pain intensity: IL-6, CSF3, and SIGLEC1 (CD169). The profound importance of IL-6 in immunoregulation and pain has been extensively studied and is reviewed elsewhere (Jenkins et al., 2021). CSF3, another member of the IL-6 superfamily of cytokines, is involved in proliferation, differentiation and function of granulocytes (Palande et al., 2013), a cell type implicated in FM pathology (Caxaria et al., 2023). Interestingly, we found that CSF3 was not only associated with reduced pain intensity, but also with reduced pressure pain sensitivity in FMS. SIGLEC1 (CD169) is a surface adhesion molecule expressed by immune cells, mainly macrophages, and is involved in innate and adaptive immunity and autoimmunity, intriguingly with indications of vitiating effect of DC169 + macrophages in autoimmune conditions such as colitis, multiple sclerosis and RA (Liu et al., 2021).

CST7 although not part of any functional clusters, stands out due to negative correlation with all four pain intensity measures as well as the FIQ score. CST7 regulates several immune processes through its inhibitory effect on cysteine cathepsins. It has inhibitory effect on the cytotoxicity of NK and T-cells and is shown to inhibit peptidases involved in antigen presentation, at least in vitro (Kos et al., 2018).

Finally, the limited overlap between the proteins differentiating FMS from HC and FMS with high vs low anti-SGC IgG suggests a heterogeneous pathophysiology within the FM group. Only six proteins met these criteria, five with no (CLSTN2, LAMA4, ITGA6, MATN2, ATP5IF1), and one with minor (LIFR) role in immunoregulatory functions. Five are linked to tissue development, maintenance and regeneration and one

(ATP5IF1) is linked to metabolism and energy homeostasis (Beck et al., 1990; García-Bermúdez & Cuezva, 2016; Hintsch et al., 2002; Khademi et al., 2023; Korpos et al., 2015; Viswanadhappalli et al., 2022).

4.4. Limitations

The major limitation is that no conclusions regarding causality can be drawn due to the cross-sectional nature of this study. Here we describe immune system-related alterations in FMS, but we are not able to answer what factors are driving those changes. An emerging research area is the possible relevance of changes in gut microbiota as a trigger of adaptive immune responses in FM (Cai et al.; Minerbi et al., 2019), but microbiota has not been analysed in the current cohort. An increased prevalence of FM has been observed among individuals with certain infectious and autoimmune conditions, although current evidence does not support a causal link (Cassisi et al., 2011; Fan et al., 2017; Lichtenstein et al., 2018; Zhao et al., 2019). In the present study we excluded subjects with autoimmune conditions (except FM) and ongoing infections. Stress could also be speculated to be a link between FM and immune system activation. Chronic stress assessed in a pain context, as well as anxiety and PTSD disorders, where physiological stress responses are central in the symptomatology, are shown to be associated with inflammatory response (Renna et al., 2018; Rivat et al., 2010). An association between stress and development of FM has been observed since long and extensively studied (Beiner et al., 2025; Beiner et al., 2023; Lyon et al., 2011; Martinez-Lavin, 2012; Van Houdenhove & Egle, 2004), although the associations between aberrant stress hormone levels and FM are inconsistent (Beiner et al., 2025; Beiner et al., 2023). However, although stress was not directly assessed in our study, “stress-related” factors such as rated anxiety, pain catastrophizing, fatigue and sleep disturbance were not associated to anti-SGC IgG levels. In order to pinpoint potential upstream drivers of immune activation, future studies could consider combining serum proteomics with gene expression analysis or immune receptor profiling, as well as extending the analysis to also include cerebrospinal fluid.

To avoid sex related physiological differences to contribute to interpretive difficulties at this stage, only female subjects were included, consequently it is unknown if the results can be generalized to men. In addition, the mean and maximum BMI values differ significantly between groups and although BMI is adjusted for in our analyses, careful BMI matching during inclusion would be preferred.

In contrast with methods using whole proteome analysis the OLINK inflammatory panel is limited to a selection of 384 proteins and provides only relative values. In addition, the data from OA and DDD subjects compared to HC were collected in a different setting than the FM data, and the mean and maximum age of the OA subjects, is considerably higher than the age of the FMS and also the DDD subjects. These circumstances constraints the scientific merit of the yet interesting comparison between the different pain conditions.

The present study is, to our knowledge, first to use a 384 OLINK inflammatory panel to analyze serum proteins in FMS and anti-SGC IgGs are a totally new concept in FM pathology. Therefore, the results of this study should be regarded as purely exploratory, and caution is warranted when interpreting the results. Complementary hypothesis-driven studies are necessary to elucidate the applicability of protein aberrations as potential diagnostic markers and treatment targets in FM.

4.5. Conclusions

Our study is the first to relate findings of elevated anti-SGC IgG to protein data in fibromyalgia (FM) and we consider it exploratory and hypothesis generating in the sense that we use a relatively large protein panel to identify proteins differing between groups and have applied a systems biology approach in interpreting the results. We identify characteristic protein interaction patterns, forming clusters aligned with different functional categories, supporting immune activation and

indicating imbalance in tissue homeostasis in FM. These results differentiated FM patients from patients suffering from nociceptive pain (osteoarthritis and degenerative disk disease) and indicate disease specific alterations. In addition, we describe that compared to HC, FMS had higher levels of anti-SGC IgG antibodies, and a positive correlation was found between the levels of these antibodies and pain intensity. A few of the proteins upregulated in the FMS with high levels of anti-SGC correlate with symptom severity while several downregulated proteins correlate inversely with pain intensity, suggesting protective roles. Furthermore, we identify specific proteins of interest, considering their association to symptom severity, their function in relation to autoimmunity or neuroplasticity, and/or their repeated appearance in high throughput protein panel studies. CD40 appears particularly relevant and warrants further attention, although several other proteins may also be of interest.

Overall, our data support the involvement of immune mechanisms and suggest imbalance in tissue homeostasis in FM but need to be reproduced in future studies. A better understanding of these mechanisms could lead to new objective diagnostic criteria identifying FMS likely to benefit from existing immunomodulatory treatments and pave the way for development of new treatment strategies for FMS.

CRedit authorship contribution statement

Karolina af Ekenstam: Writing – original draft, Methodology, Data curation, Conceptualization. **Joana Menezes:** Writing – review & editing, Methodology, Formal analysis. **Jenny E. Jakobsson:** Writing – review & editing, Formal analysis. **Helene Silverstein:** Writing – original draft, Data curation. **Emerson Krock:** Writing – review & editing, Methodology, Funding acquisition, Formal analysis. **Jeanette Tour:** Writing – review & editing, Methodology, Investigation, Data curation. **Katalin Sandor:** Writing – review & editing, Methodology, Formal analysis. **Alexandra Kuliszkiwicz:** Writing – review & editing, Methodology, Formal analysis. **Matthew Hunt:** Writing – review & editing, Methodology, Formal analysis. **Kim Kultima:** Writing – review & editing, Resources, Methodology, Funding acquisition, Formal analysis. **Macarena Tejos-Bravo:** Formal analysis, Investigation, Visualization, Writing – review & editing. **Camilla I. Svensson:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Eva Kosek:** Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Web references

The Human Protein Atlas.
<http://www.proteinatlas.org>

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bbi.2025.106185>.

Data availability

The authors do not have permission to share data.

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